UNIVERSITY OF BATNA 2

Faculty of Mathematics and Computer Science

Department of Computer Science

4th year Engineers: Security

2025/2026

NETWORK SECURITY

Practical Work 7: SSL/TLS INSPECTION REPORT

Team Members

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Deadline

26 April 2026

Objectives

In this lab, you will:

- Understand HTTPS/TLS operation
- Configure a TLS inspection proxy
- Install and trust a custom root CA
- Intercept HTTPS traffic
- Analyze intercepted credentials
- Extract downloaded files from traffic

- Analyze suspicious files using VirusTotal

Required Environment

- 1 Linux VM (Kali / Ubuntu)
- 1 Windows 7 VM or your host machine
- mitmproxy installed
- Web Browser (Firefox)

Tasks to Perform : follow the steps in the documentation of the lab I sent you last week

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Part 1 – Proxy Configuration

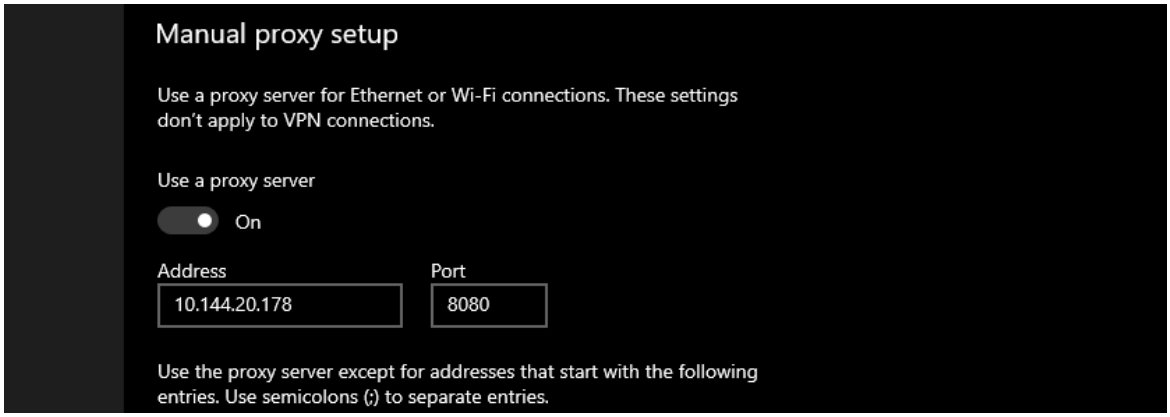
you must:

- **Start mitmproxy on Linux**

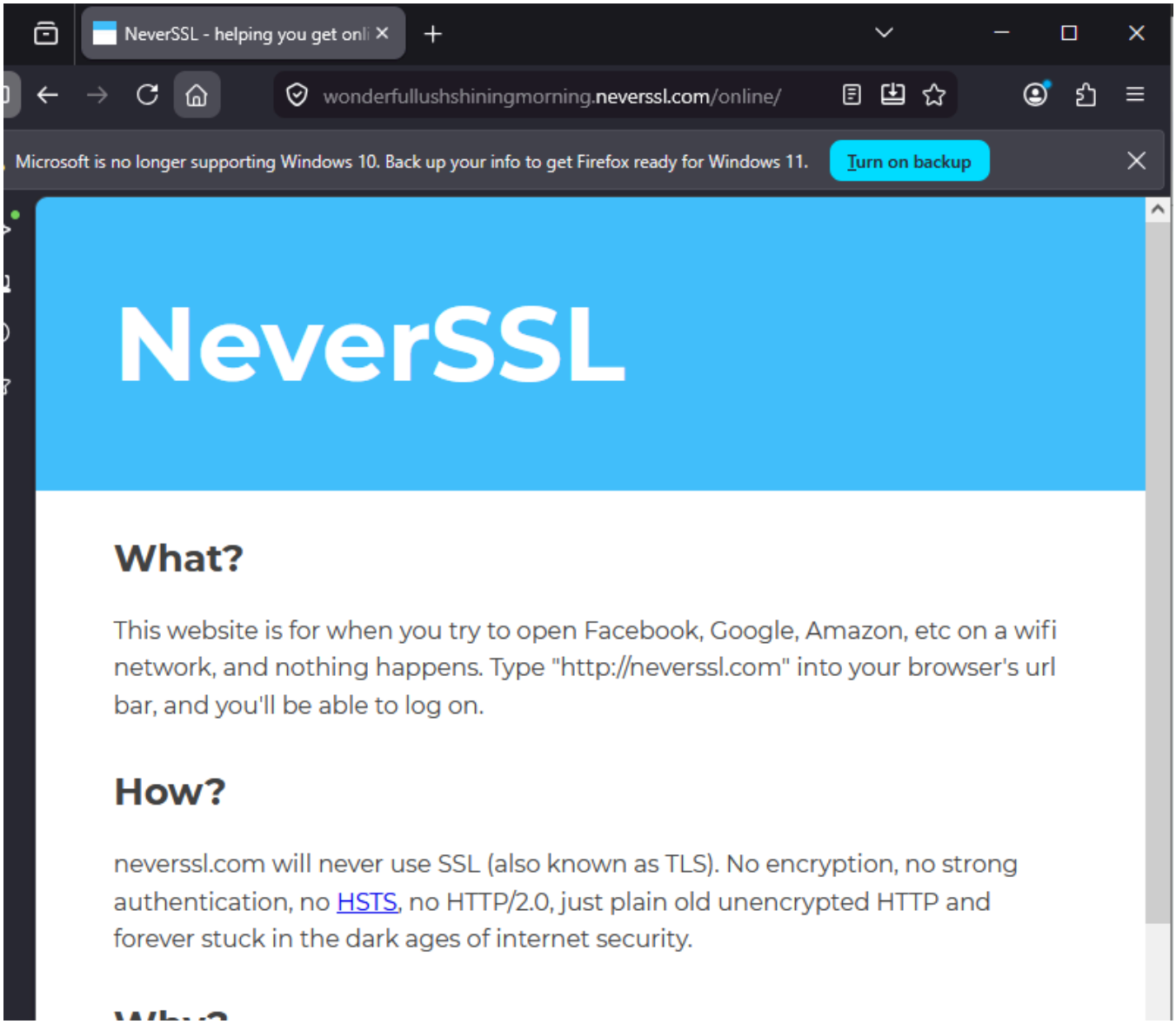
```
┌──(kali㉿kali)-[~]
└─$ sudo ufw disable
Firewall stopped and disabled on system startup
┌──(kali㉿kali)-[~]
└─$ sudo apt install mitmproxy -y
mitmproxy is already the newest version (12.2.1-0kali1).
mitmproxy set to manually installed.
The following packages were automatically installed and are no longer required:
  libdaq3  libpython3.12-minimal  libpython3.12-stdlib  libpython3.12t64  oinkmaster
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2201
```

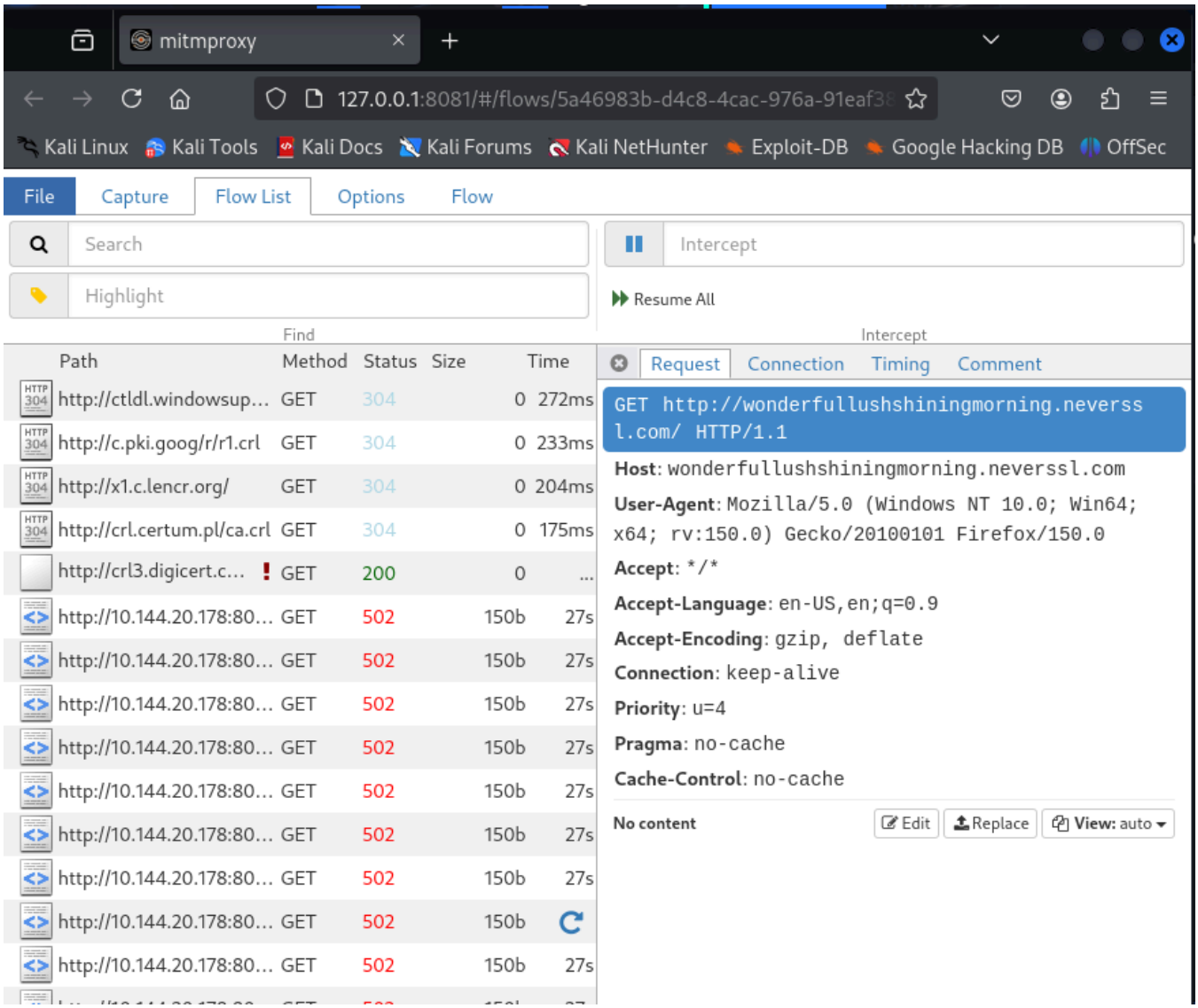
```
┌──(kali㉿kali)-[~]
└─$ sudo mitmweb --listen-host 0.0.0.0 --listen-port 8080
[18:19:01.223] HTTP(S) proxy listening at *:8080.
[18:19:01.225] Web server listening at http://127.0.0.1:8081/?token=970705bc2c00f2c279a3eaadd3f049
70
[18:19:31.548][10.144.20.240:49996] client connect
[18:19:31.731][10.144.20.240:49996] server connect ctldl.windowsupdate.com:80 (146.75.122.172:80)
[18:19:31.977][10.144.20.240:49996] server connect c.pki.goog:80 (142.250.181.99:80)
[18:19:32.197][10.144.20.240:49996] server connect x1.c.lencr.org:80 (2.20.33.120:80)
[18:19:32.276][10.144.20.240:49996] server disconnect x1.c.lencr.org:80 (2.20.33.120:80)
[18:19:32.277][10.144.20.240:49996] client disconnect
[18:19:32.277][10.144.20.240:49996] server disconnect ctldl.windowsupdate.com:80 (146.75.122.172:8
0)
[18:19:32.284][10.144.20.240:49996] server disconnect c.pki.goog:80 (142.250.181.99:80)
[18:19:32.285][10.144.20.240:50120] client connect
[18:19:32.396][10.144.20.240:50120] server connect crl.certum.pl:80 (23.53.1.139:80)
[18:19:32.466][10.144.20.240:50120] server disconnect crl.certum.pl:80 (23.53.1.139:80)
[18:19:32.469][10.144.20.240:50120] client disconnect
[18:19:32.469][10.144.20.240:50121] client connect
[18:19:32.596][10.144.20.240:50121] server connect crl3.digicert.com:80 (23.11.40.157:80)
[18:19:39.138][10.144.20.240:50240] client connect
[18:19:39.157][10.144.20.178:47740] client connect
[18:19:39.157][10.144.20.240:50240] server connect 10.144.20.178:8080
[18:19:39.163][10.144.20.178:47750] client connect
[18:19:39.164][10.144.20.178:47740] server connect 10.144.20.178:8080
[18:19:39.166][10.144.20.178:47756] client connect
[18:19:39.174][10.144.20.178:47750] server connect 10.144.20.178:8080
[18:19:39.177][10.144.20.178:47762] client connect
[18:19:39.179][10.144.20.178:47756] server connect 10.144.20.178:8080
[18:19:39.193][10.144.20.178:47766] client connect
```

- **Configure Windows proxy settings**



- **Verify connectivity through the proxy**


NeverSSL - helping you get onli
wonderfullushshiningmorning.neverssl.com/online/
Microsoft is no longer supporting Windows 10. Back up your info to get Firefox ready for Windows 11.
Turn on backup
NeverSSL
What?
This website is for when you try to open Facebook, Google, Amazon, etc on a wifi network, and nothing happens. Type "http://neverssl.com" into your browser's url bar, and you'll be able to log on.
How?
neverssl.com will never use SSL (also known as TLS). No encryption, no strong authentication, no HSTS, no HTTP/2.0, just plain old unencrypted HTTP and forever stuck in the dark ages of internet security.



Q1: Why must the client configure a proxy?

The client must configure a proxy so that all network traffic is redirected through the interception system (mitmproxy). This allows the proxy to capture, inspect, and analyze HTTP/HTTPS requests and responses for security monitoring and TLS inspection

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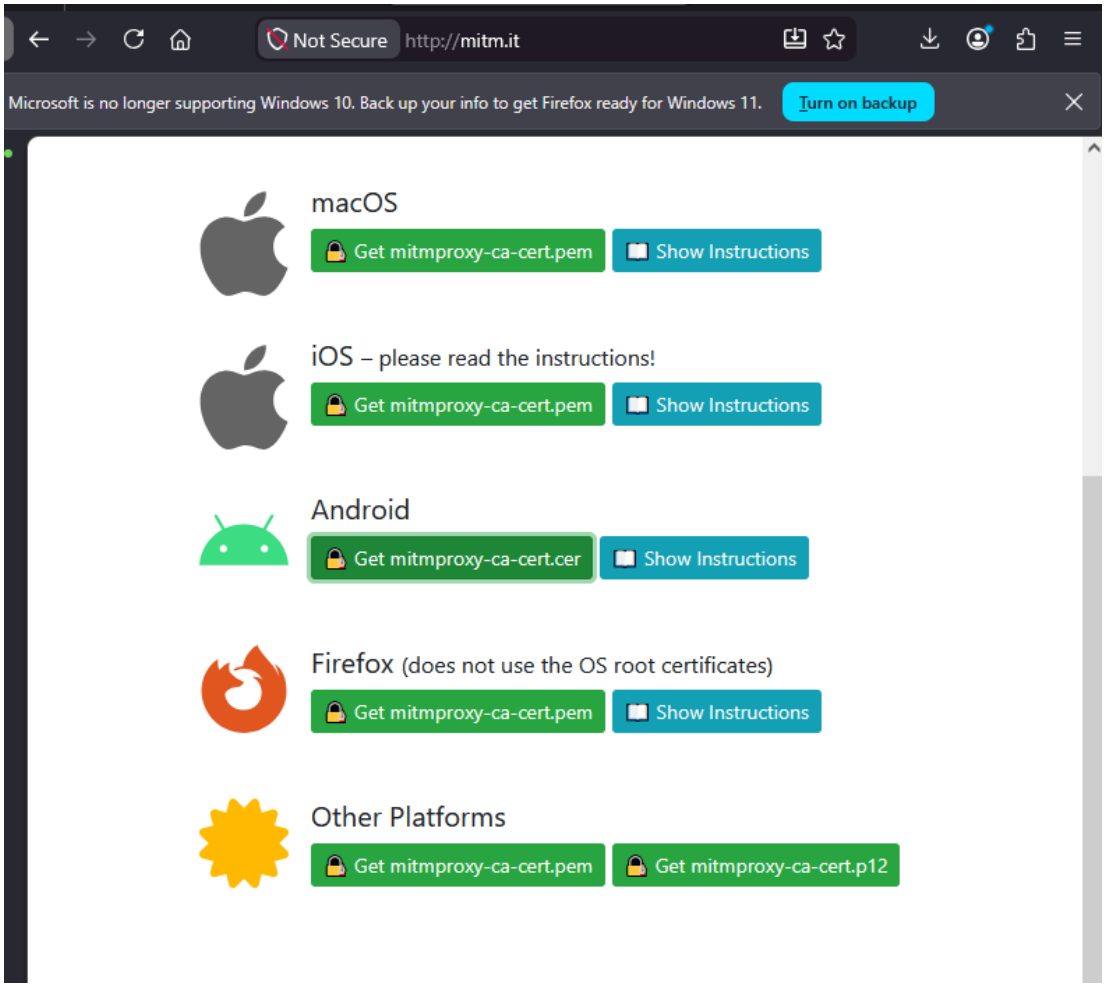
Part 2 – Root Certificate Installation

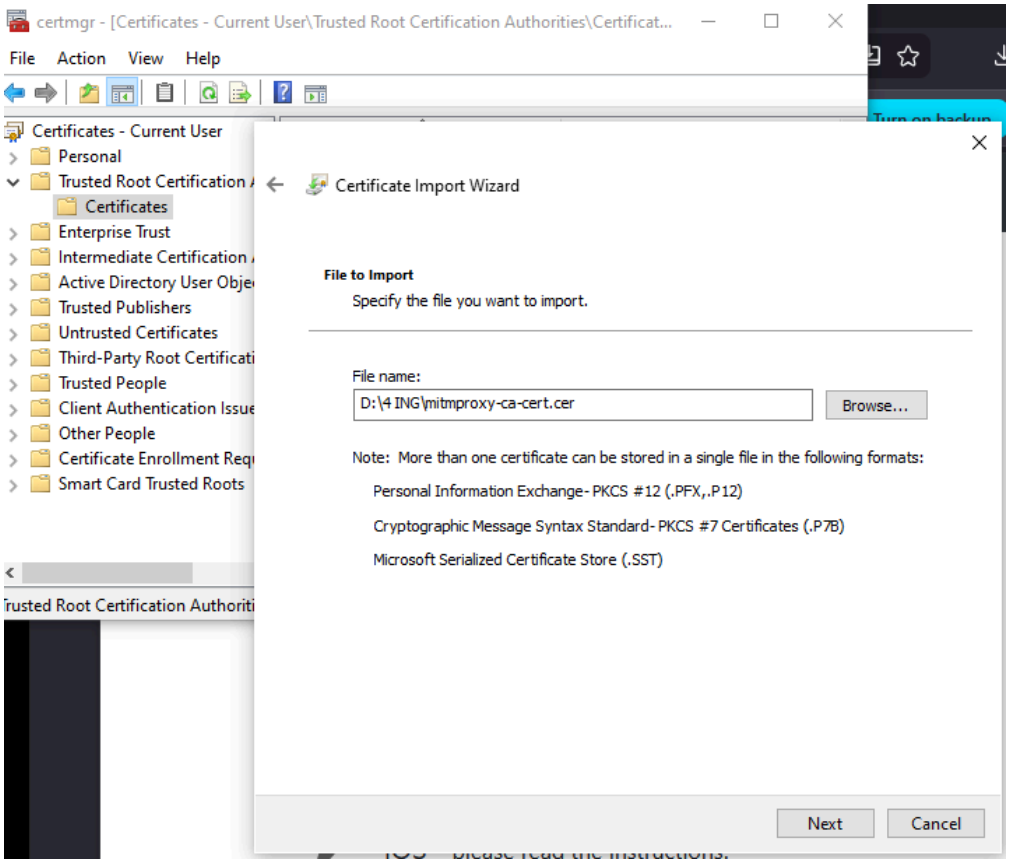
you must:

- ***Download the mitmproxy certificate from http://mitm.it***

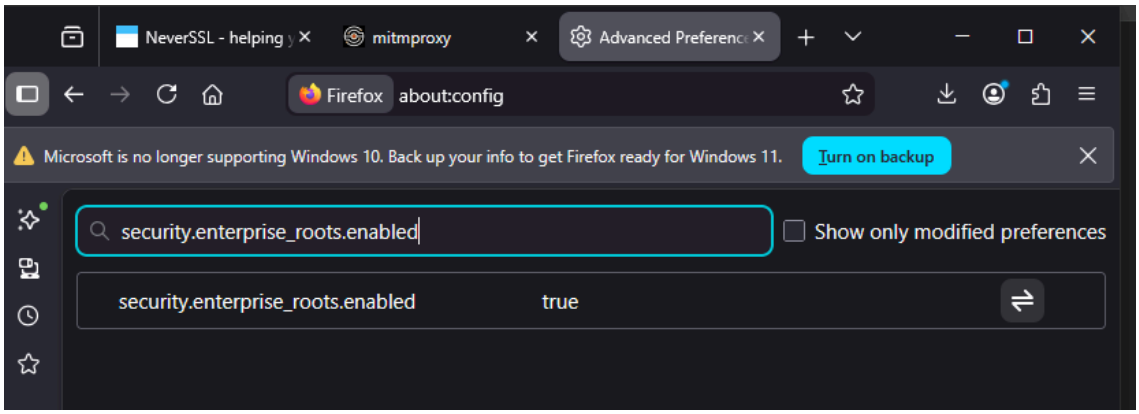


Click on Certaficate of Abdroid for download

- ***Install it into Trusted Root Certification Authorities in your local machine***



- ***Enable Firefox enterprise certificate trust***



1. *Explain:*

- ***Why is this certificate required?***

The mitmproxy root certificate is required to allow the browser to trust the proxy during HTTPS interception. It enables mitmproxy to generate and sign certificates for websites so encrypted traffic can be decrypted, inspected, and re-encrypted transparently

- ***What happens if it is not installed*?**

  If the certificate is not installed, the browser will not trust the proxy. HTTPS connections will show security warnings or fail, and encrypted traffic cannot be intercepted or analyzed

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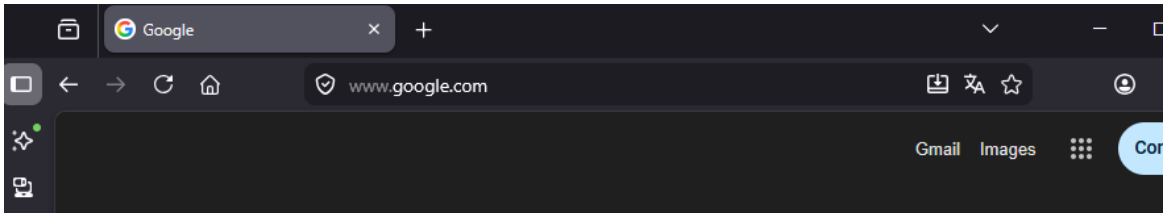
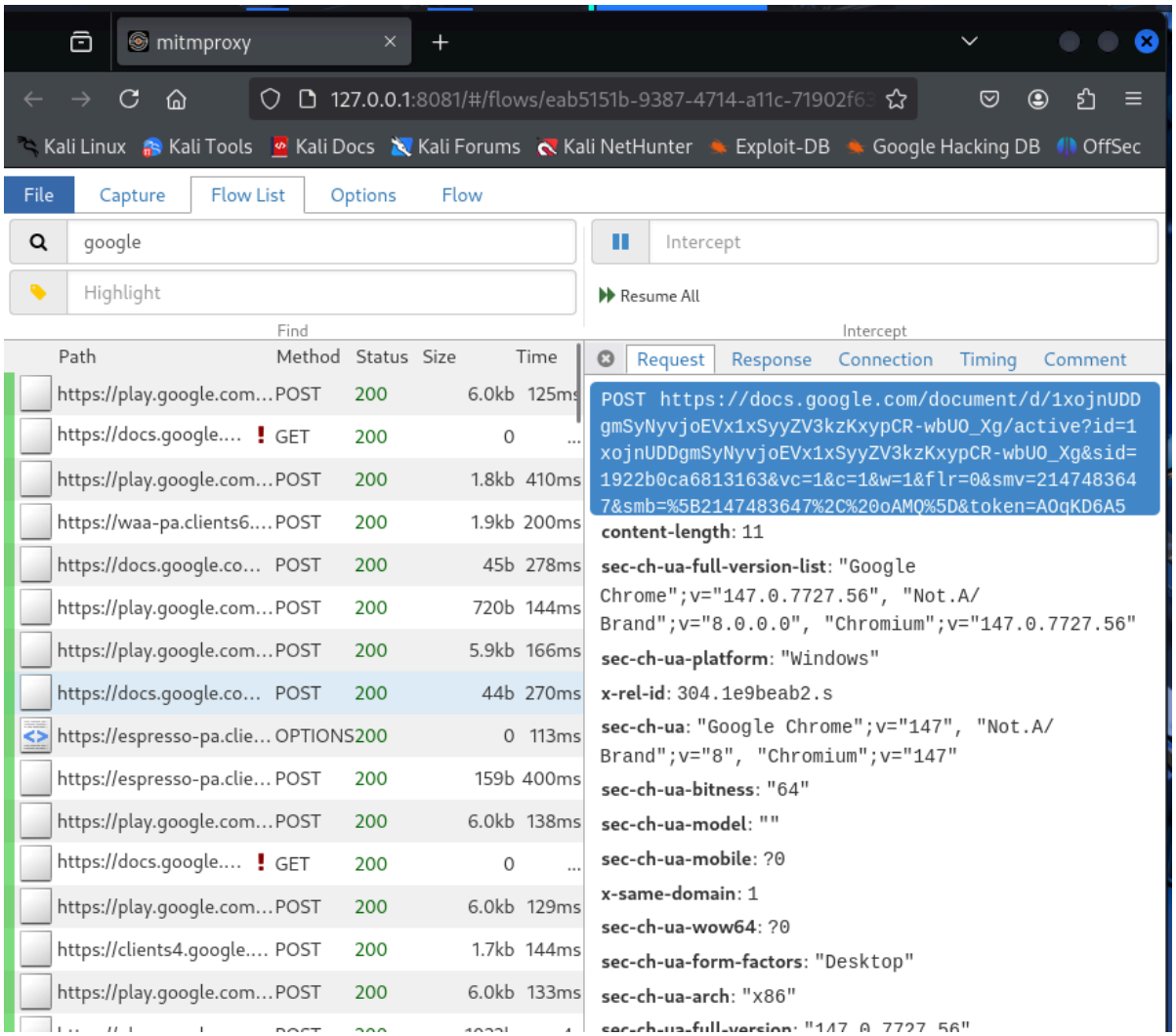
Part 3 – TLS Interception Verification

you must:

- ***Browse to https://google.com***



- ***Confirm HTTPS interception in mitmproxy***



- ***Inspect certificate details***

Certificate Viewer: www.google.com

General | Details

Issued To

Common Name (CN)	www.google.com
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	mitmproxy
Organization (O)	mitmproxy
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Thursday, April 23, 2026 at 12:29:38 AM
Expires On	Sunday, April 25, 2027 at 12:29:38 AM

SHA-256 Fingerprints

Certificate	b3e172ef32edc455834332235a04a024d322dda03f853d1dcbcdd9dd46903f17
Public Key	6def7001bd519be37187bc4c24d31eec88506a92f0e40fe3b72e8b5bce3bc428

Q2: Why does the browser trust the intercepted connection?

The browser trusts the intercepted connection because the mitmproxy root certificate is installed in the trusted store. This allows mitmproxy to generate valid certificates for HTTPS websites, so the browser considers the connection secure

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Part 4 – Credential Interception

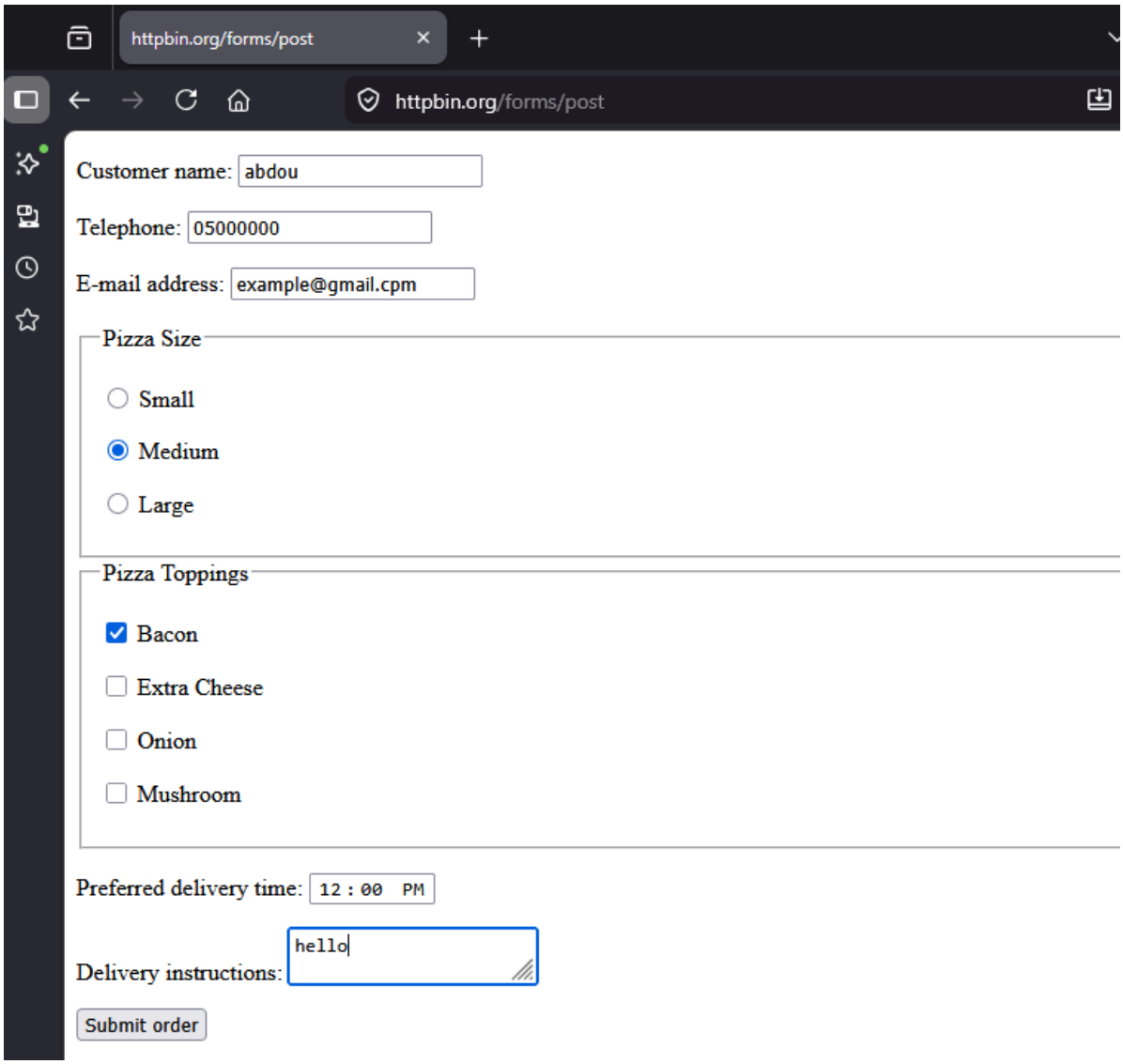
you must:

- ***Visit https://httpbin.org/forms/post***
- ***Submit data of the team members (all the names one after the other and the email of the team leader in the corresponding forms)***
- ***Locate the POST request in mitmproxy***
- ***Inspect request body***

Deliverables

Include:

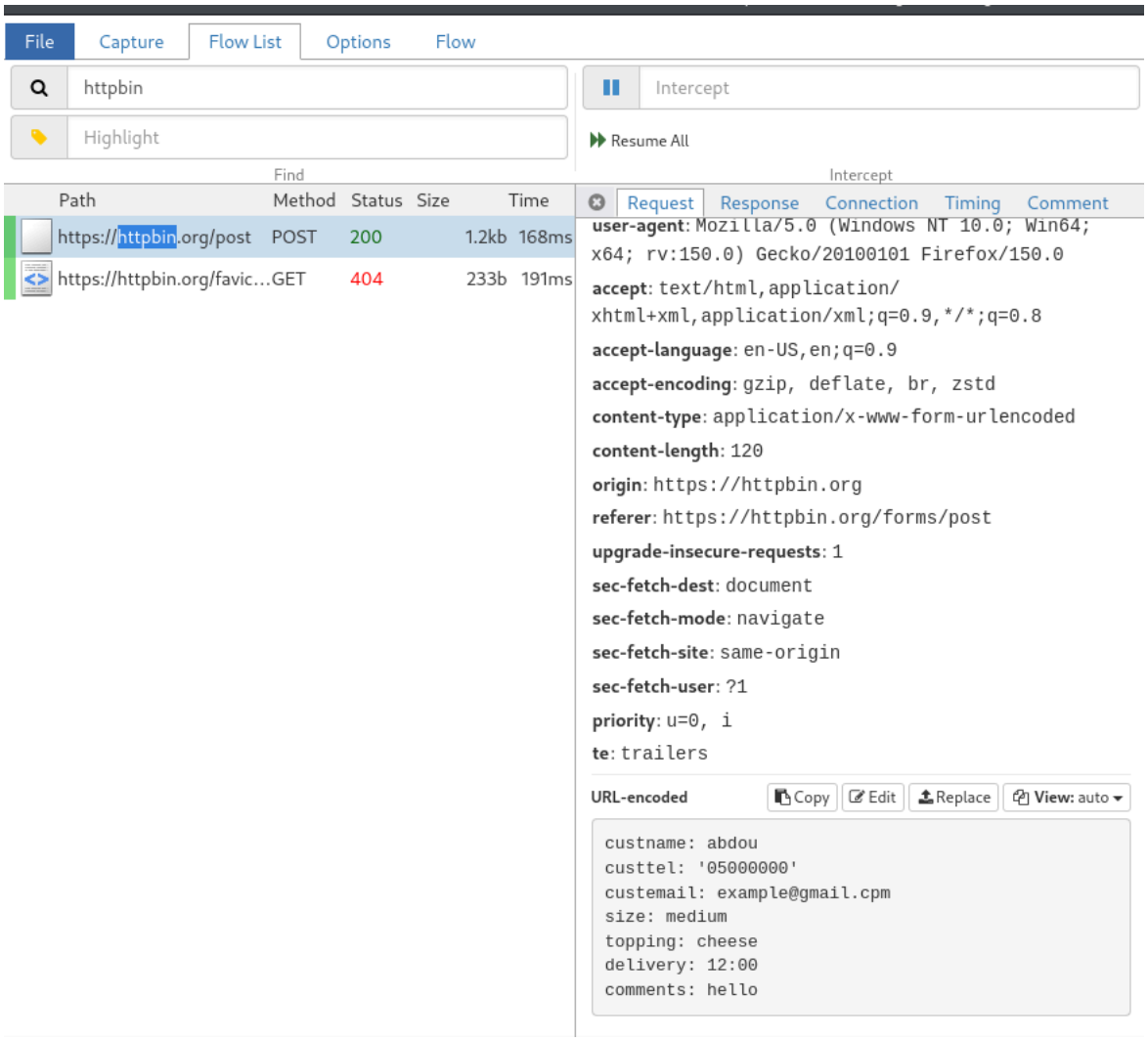
1. ***Screenshot of POST request***


httpbin.org/forms/post
httpbin.org/forms/post
Customer name: abdou
Telephone: 05000000
E-mail address: example@gmail.cpm
Pizza Size
Small
Medium
Large
Pizza Toppings
Bacon
Extra Cheese
Onion
Mushroom
Preferred delivery time: 12 : 00 PM
Delivery instructions: hello
Submit order

JSON Raw Data Headers

Save Copy Collapse All Expand All Filter JSON


```
  args:                          {}
  data:                          ""
  files:                         {}
▼ form:
    comments:                    "hello"
    custemail:                   "example@gmail.cpm"
    custname:                    "abdou"
    custtel:                     "05000000"
    delivery:                    "12:00"
    size:                        "medium"
    topping:                     "cheese"
▼ headers:
    Accept:                      "text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8"
    Accept-Encoding:             "gzip, deflate, br, zstd"
    Accept-Language:             "en-US,en;q=0.9"
    Content-Length:              "120"
    Content-Type:                "application/x-www-form-urlencoded"
    Host:                        "httpbin.org"
    Origin:                      "https://httpbin.org"
    Priority:                    "u=0, i"
    Referer:                     "https://httpbin.org/forms/post"
    Sec-Fetch-Dest:              "document"
    Sec-Fetch-Mode:              "navigate"
    Sec-Fetch-Site:              "same-origin"
    Sec-Fetch-User:              "?1"
    Upgrade-Insecure-Requests:   "1"
    User-Agent:                  "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:150.0) Gecko/20100101 Firefox/150.0"
    X-Amzn-Trace-Id:             "Root=1-69ebfb33-121d33c407a723ad15d0857d"
  json:                          null
  origin:                        "41.200.43.55"
  url:                           "https://httpbin.org/post"
```



So can itercept Credentials

2. ***Answer:***

Q3: ***Why are credentials visible despite HTTPS?***

Credentials are visible because TLS inspection decrypts the HTTPS traffic at the proxy. Since the mitmproxy certificate is trusted, the proxy can access and display the data in plaintext before re-encrypting it.

Part 5 – Suspicious File Download Analysis

you must:

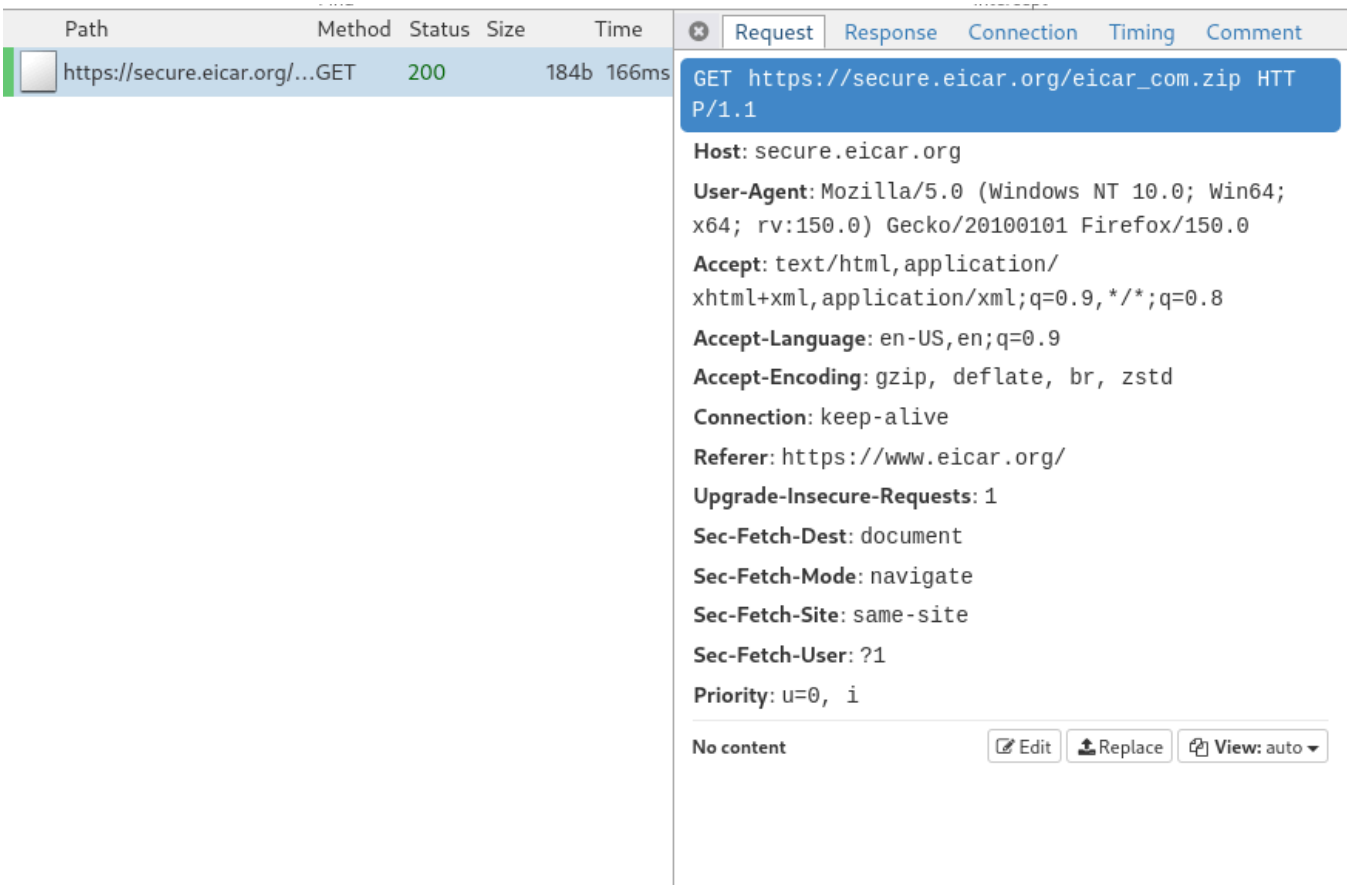
- **Download the malware zip from https://www.eicar.org/download-anti-malware-testfile/**

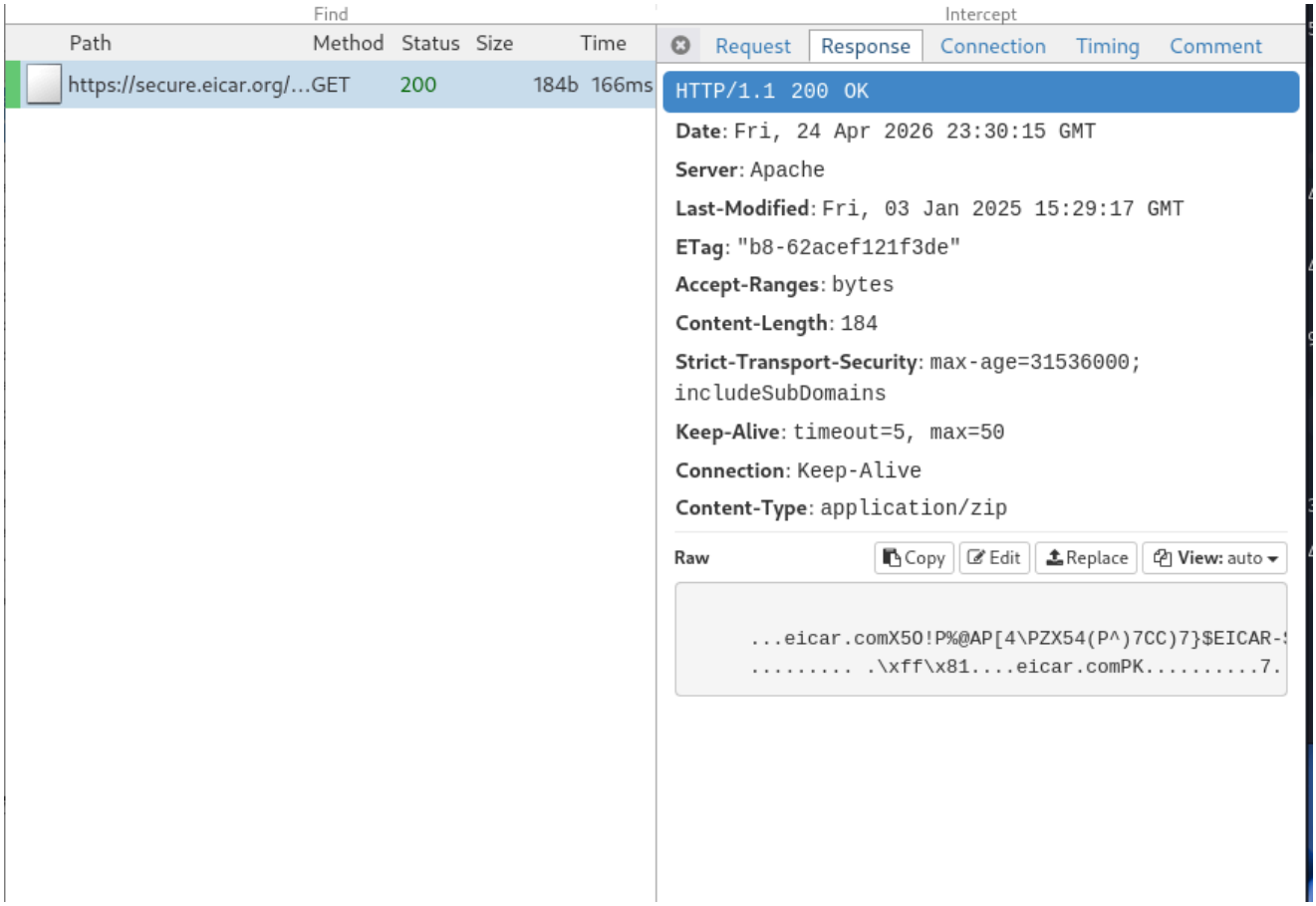
- **Locate the file download in mitmproxy**
- **Coose the Response to find the bytecode**
- **Extract the downloaded file using Save Body**

- **Save it locally as**

intercepted.zip Deliverables

Include:





Use commands for save zip file

```
┌──(kali㉿kali)-[~]
└─$ sudo mitmproxy --listen-host 0.0.0.0 --listen-port 8080
```

```
File  Actions  Edit  View  Help
Flow Details
2026-04-24 19:57:53 GET https://secure.eicar.org/eicar_com.zip
                    ← 200 OK application/zip 184b 156ms
        Request                 Response                    Detail
Date:                           Fri, 24 Apr 2026 23:57:54 GMT
Server:                         Apache
Last-Modified:                  Fri, 03 Jan 2025 15:29:17 GMT
ETag:                           "b8-62acef121f3de"
Accept-Ranges:                  bytes
Content-Length:                 184
Strict-Transport-Security:      max-age=31536000; includeSubDomains
Keep-Alive:                     timeout=5, max=50
Connection:                     Keep-Alive
Content-Type:                   application/zip
Raw                                                                  [m:auto]
PK..
.....\xe0\x98\xb8(<\xcfQhD ... D ... ? ... eicar.comX5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS
TEST-FILE!$H+H*PK....
.....\xe0\x98\xb8(<\xcfQhD ... D ... ?......... .\xff\x81....eicar.comPK..........7 ... k.....

⇩  [6/86]                                                            [*:8080]
Flow:    e Edit       D Duplicate   r Replay     x Export     d Delete     b Save body
Proxy:   ? Help       q Back        E Events     O Options    i Intercept  f Filter
```

And after cliwk on w and write name of file

```
Flow Details
2026-04-24 19:57:53 GET https://secure.eicar.org/eicar_com.zip
                    ← 200 OK application/zip 184b 156ms
        Request                Response                Detail
Date:                          Fri, 24 Apr 2026 23:57:54 GMT
Server:                        Apache
Last-Modified:                 Fri, 03 Jan 2025 15:29:17 GMT
ETag:                          "b8-62acef121f3de"
Accept-Ranges:                 bytes
Content-Length:                184
Strict-Transport-Security:     max-age=31536000; includeSubDomains
Keep-Alive:                    timeout=5, max=50
Connection:                    Keep-Alive
Content-Type:                  application/zip
Raw                                                                  [m:auto]
PK..
.....\xe0\x98\xb8(<\xcfQhD ... D ... ? ... eicar.comX5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS
TEST-FILE!$H+H*PK....
.....\xe0\x98\xb8(<\xcfQhD ... D ... ?......... .\xff\x81....eicar.comPK..........7 ... k.....

⇩  [6/357]                                                           [*:8080]
Alert: [20:02:38.931] Saved 1 flows.
```

```
┌──(kali㉿kali)-[~]
└─$ ls intercepted.zip
intercepted.zip
```

1. ***Answer:***

Q4: What does the intercepted content represent?

The intercepted content represents an EICAR test file inside a ZIP archive. It is a harmless standard file used to simulate malware and test antivirus and security inspection systems

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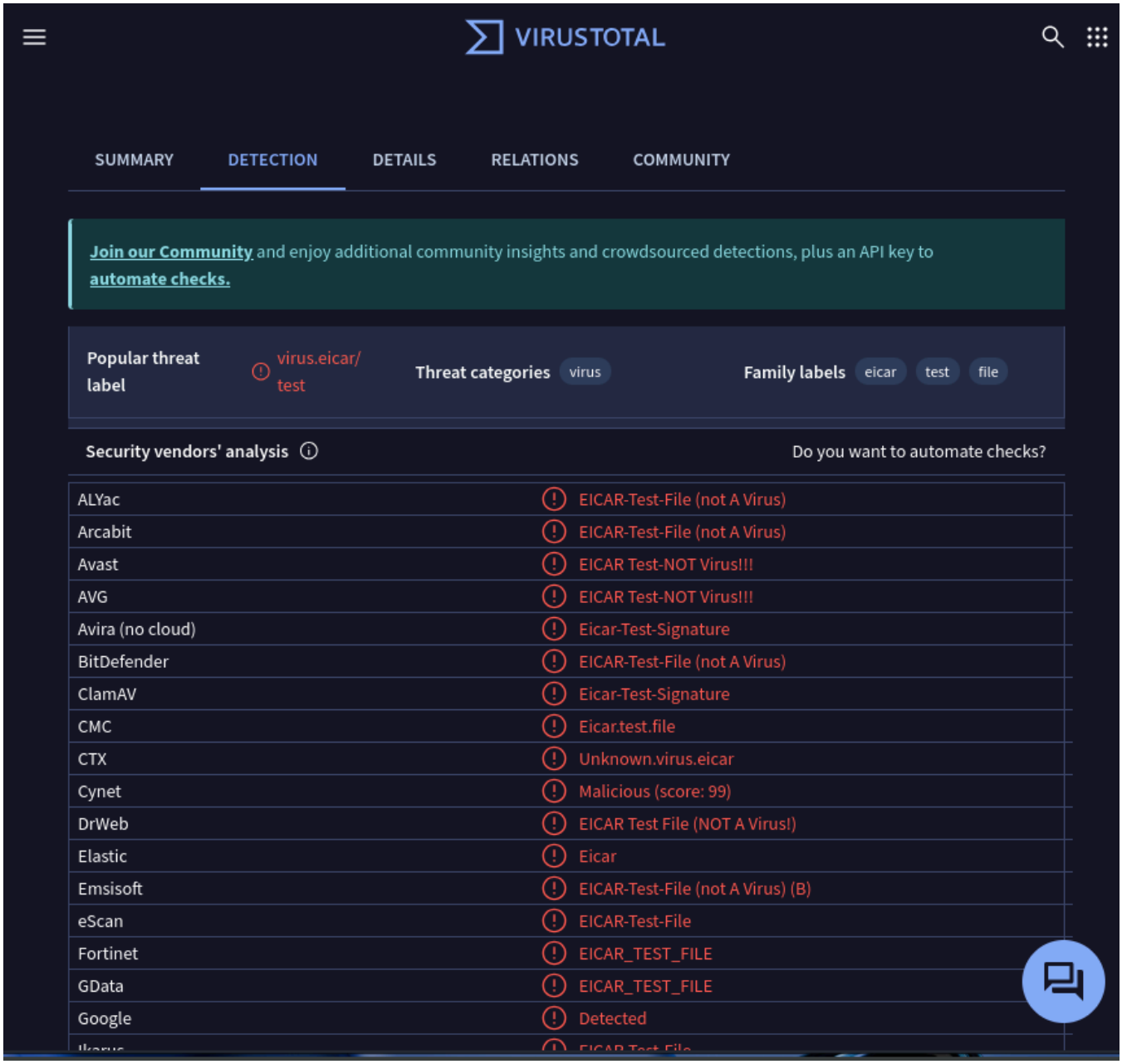
Part 6 – Malware Analysis

Students must:

- **Upload extracted file to VirusTotal**

- **Analyze detection**

The file is detected by multiple antivirus engines as an EICAR test file. It is not a real malware but a standardized file used to test security systems. VirusTotal confirms that security vendors recognize it as a simulation of malicious content



1. **Q5:** ***Although the malicious file was intercepted and inspected in plaintext through TLS inspection, the user was still able to download it. What type of enterprise security appliance could be deployed inline with the TLS inspection system to block malicious downloads in real time? Explain how it would enhance the protection provided by TLS inspection.***

   Answer:

   A Next-Generation Firewall (NGFW) with Intrusion Prevention System (IPS) can be deployed inline with TLS inspection. It analyzes decrypted traffic in real time and blocks malicious downloads before they reach the user. While TLS inspection only decrypts and exposes traffic, the NGFW/IPS adds active threat prevention. This provides real-time blocking of malware and enhances overall security

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Required Report Structure

For each part above, include:

1. Objective of the step
2. Configuration / Commands Used
3. Screenshots / Evidence
4. Technical Explanation
5. Security Analysis
6. Answers to Questions
7. Issues Encountered / Troubleshooting